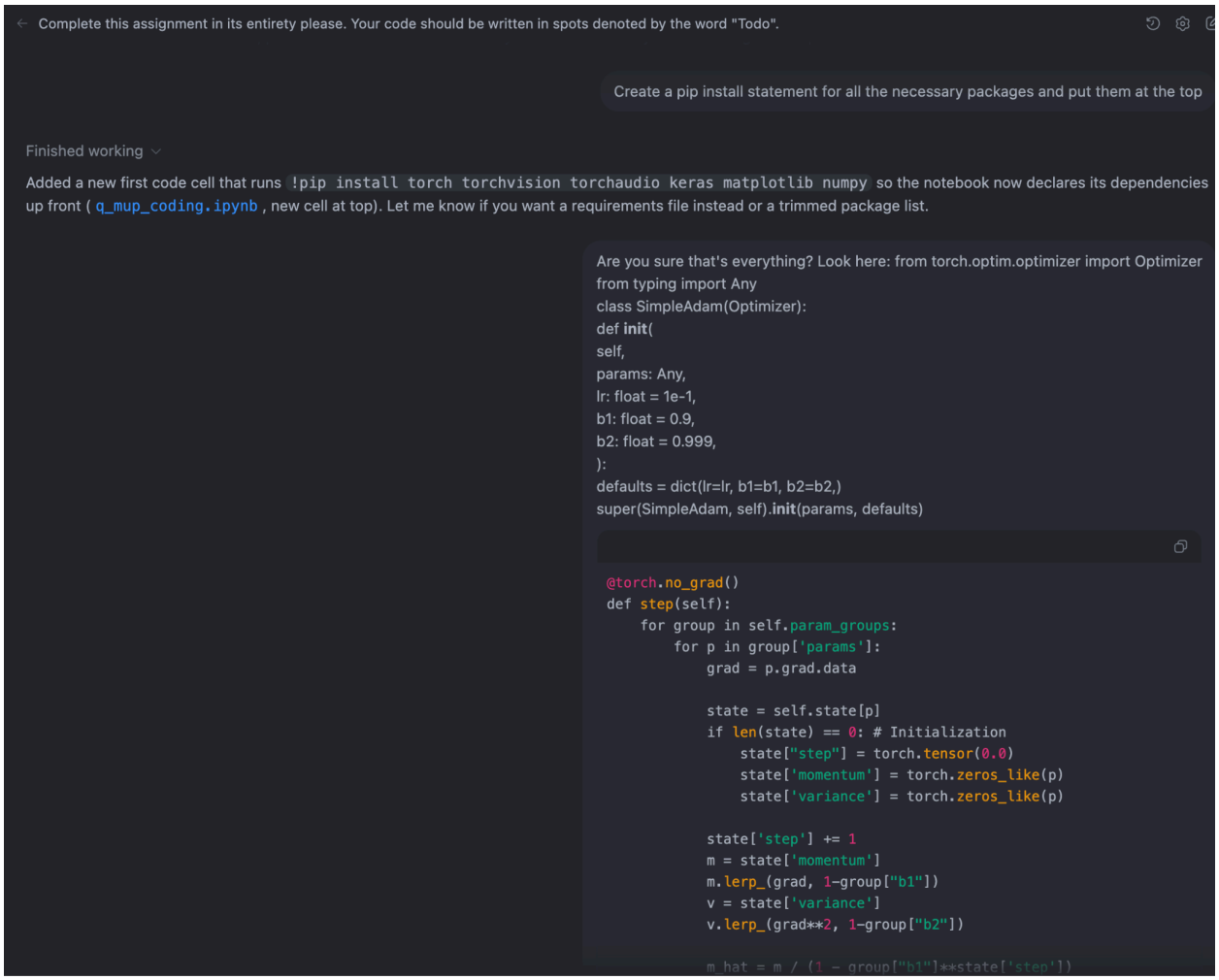
There are two main components in this assignment that I wanted Codex to solve: the coding portions and the answer portions. I first tried asking it to solve and implement everything at once, without any specificity towards examining figures/images.



The answers and code were filled in and were seemingly correct, but lacked some specificity.

CODEX

Complete this assignment in its entirety please. Your code should be written in spots denoted by the word "Todo".

Complete this assignment in its entirety please. Your code should be written in spots denoted by the word "Todo".

Finished working

Key Changes

- Added a second invocation of `train_one_step` with wider early layers to recreate the activation-delta plot after changing widths, satisfying part (a)'s plotting request (`q_mup_coding.ipynb (line 194)`).
- Rebuilt `train_one_step_matrices` to capture Frobenius, spectral, and RMS-RMS induced norms (with readable layer labels) so the weight-update analysis in part (b) now works end-to-end (`q_mup_coding.ipynb (line 229)`).
- Implemented the muP-aware optimizer by scaling each parameter's learning rate by its fan-in, with the requested bias/input/output fudge factors, yielding uniform activation deltas across widths (`q_mup_coding.ipynb (line 308)`).
- Added graph-level muP parameterization by normalizing each hidden layer's pre-activations by `sqrt(fan_in)` inside `ScaledMLP` , enabling Adam to reproduce the uniform curves without optimizer changes (`q_mup_coding.ipynb (line 404)`).
- Filled in the simplified Shampoo update by orthogonalizing dense updates via SVD, so you can now contrast Frobenius vs. spectral norms for Adam vs. Shampoo (`q_mup_coding.ipynb (line 556)`).

Notes

- Tests/plots were not rerun here (not requested); please execute the relevant notebook cells to regenerate figures with the new logic.

Can you fill in all the the "Answer:" after the questions as well please

Finished working

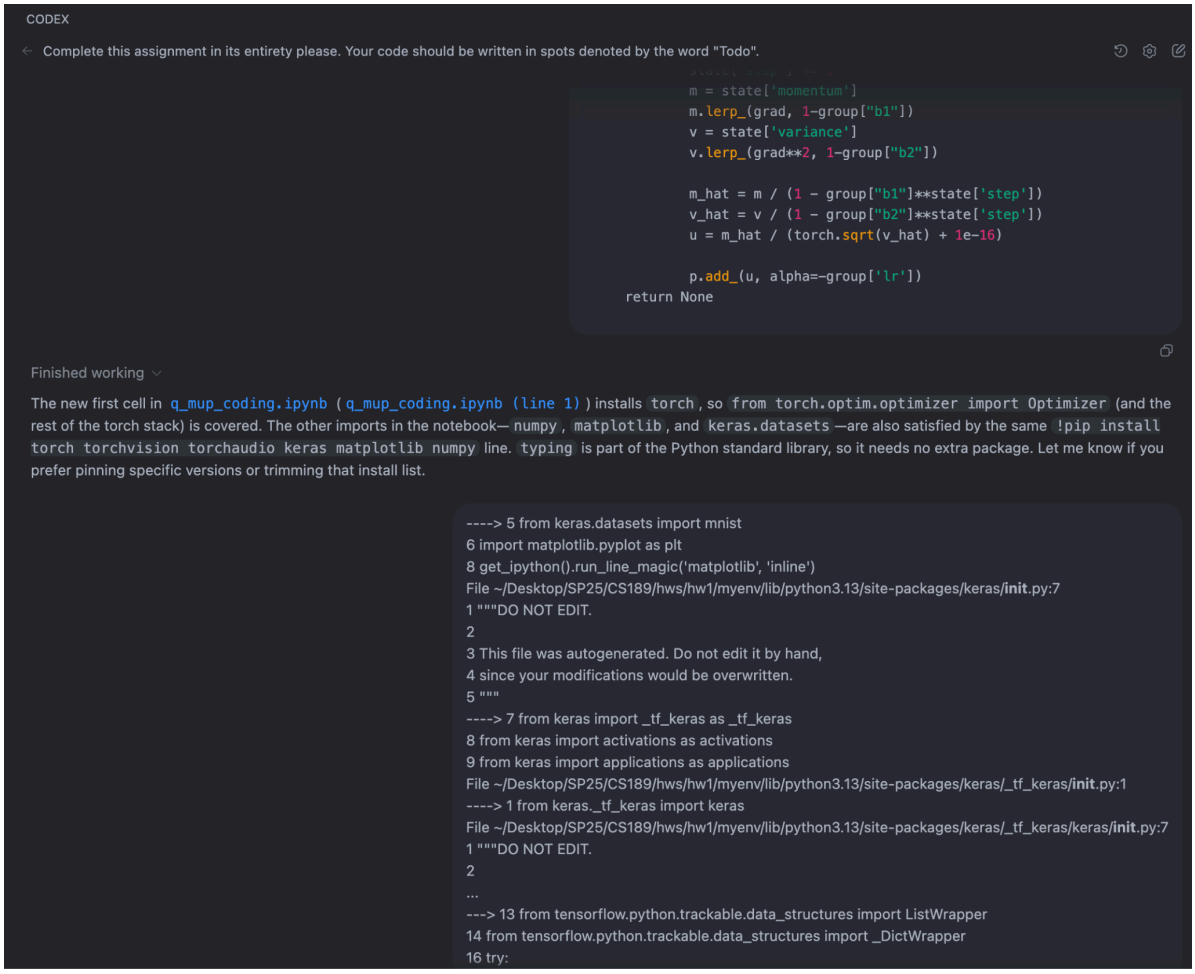
Added narrative responses for every "Answer" prompt:

- Documented the network dimensions and behavior of RMS activation deltas before and after width changes (`q_mup_coding.ipynb (line 182)`).
- Explained why the RMS–RMS induced norm best matches activation changes (`q_mup_coding.ipynb (line 219)`).
- Clarified how scaling a layer's output is equivalent to adjusting its learning rate under Adam/SignGD (`q_mup_coding.ipynb (line 387)`).
- Described the learning-rate sweep behavior for plain Adam vs. muP scaling (`q_mup_coding.ipynb (line 436)`).
- Summarized the Frobenius vs. spectral norm relationship for Adam and Shampoo updates (`q_mup_coding.ipynb (line 614)`).

No notebook cells were executed; please re-run the relevant ones in your environment if you want to regenerate plots.

Create a pip install statement for all the necessary packages and put them at the top

I decided to install all the necessary dependencies so that this specificity issue could be fixed.



After this following prompt, the answers were detailed, correct, and specific. Not much of a problem for Codex at all, though this assignment is certainly more terse than others.